

Reducing Customer Churn at Meridian Retail Group

Business & Technical Report

End-to-End Data Science Investigation

Customer Churn Capstone Project

1. Executive Summary

Meridian Retail Group is a multi-channel retailer operating across Electronics, Home & Kitchen, Apparel, Sports & Outdoors, and Beauty & Personal Care. This project investigates customer churn using customer, product, order, order-item and support-ticket data.

The workflow covered data cleaning, exploratory analysis, RFM feature engineering, support-ticket aggregation, supervised machine learning, model interpretation and API deployment. Churn was defined using a 180-day inactivity rule based on completed orders.

The final customer population contained 6,090 customers: 4,095 churned (67.24%) and 1,995 active (32.76%). Multiple models were evaluated and a tuned Random Forest was retained as the final model. On the held-out test set it achieved 0.6576 accuracy, 0.6273 balanced accuracy, 0.6234 Macro F1 and 0.6649 ROC-AUC.

The model should be treated as a customer-prioritisation tool rather than a perfect classifier. Business deployment should explicitly consider the relative financial cost of missed churners and unnecessary retention interventions.

2. Business Understanding

2.1 Business problem

Meridian Retail Group identified a need to understand lower repeat-purchase behaviour and the role of customer-support activity in customer retention. The project therefore focuses on identifying customers whose behaviour is consistent with churn and translating the analysis into actionable retention insight.

2.2 Objectives

- Define a reproducible churn target using the 180-day inactivity rule.
- Identify behavioural and customer characteristics associated with churn.
- Measure customer engagement using RFM analysis.
- Investigate support-ticket activity, satisfaction and resolution time.
- Build and compare multiple predictive models while avoiding target leakage.
- Tune and save a final model and expose it through a validated API.

2.3 Key data questions

- How many customers are churned under the 180-day definition?
- How do recency, frequency and monetary value differ by churn status?
- How does support activity vary with churn?
- Which features contribute most to model predictions?
- What are the business consequences of false positives and false negatives?

3. Data and Cleaning

Dataset	Initial records	Purpose
Customers	6,090	Customer attributes and membership information
Products	220	Product, category and pricing information
Orders	28,297	Transaction-level information
Order items	73,505	Products and quantities within orders
Support tickets	5,200	Customer-support interactions

3.1 Cleaning decisions

- Invalid DOB values such as 1899-01-01 and future DOB values such as 2031-05-04 were converted to missing.
- Missing DOB values were retained as missing rather than inventing dates.
- Twenty-five exact duplicate orders were removed.
- Forty exact duplicate order-item records were removed.
- Negative order-item quantities were investigated as invalid transaction values.
- Repeated order IDs in order items were retained because an order can legitimately contain multiple products.
- Observed order discounts between 0% and 25% were retained.
- Product unit-cost outliers were retained for investigation because they may represent legitimate variation or cost/price inconsistencies.
- Support-ticket dates were converted, invalid satisfaction values outside 1–5 were treated as missing, and negative resolution times were treated as missing.
- Scaling was deferred to modelling and preprocessing was handled within the model pipeline.

4. Churn Definition and Feature Engineering

4.1 Target

At the defined snapshot, a customer was classified as Churned when there was no completed order during the preceding 180 days, including customers with no qualifying order history. Otherwise the customer was classified as Active.

The final target contained 4,095 Churned customers (67.24%) and 1,995 Active customers (32.76%).

4.2 RFM features

- Recency_Days — days since the most recent completed purchase.
- Frequency — number of qualifying purchases.
- Monetary_Total — total customer monetary value.
- Monetary_Average — average transaction monetary value.

4.3 Support features

- Support_Ticket_Count
- Average_Satisfaction
- Average_Resolution_Time

4.4 Leakage prevention

Recency_Days was excluded from the final ML feature set because the churn target is itself defined from recent purchase inactivity. Including it would allow the model to reproduce the target rule rather than learn an independent predictive relationship.

5. Exploratory Data Analysis

5.1 RFM findings

Mean recency was approximately 89 days for active customers versus approximately 516 days for churned customers. Active customers averaged about 4.79 purchases compared with 4.25 for churned customers. Mean total monetary value was approximately 7,424.51 for active customers versus 6,511.76 for churned customers.

5.2 Support findings

Customers with 5 or more support tickets had an observed churn rate of approximately 81.82%, compared with approximately 66.61% among customers with no tickets. Customers with 3–4 tickets had an observed churn rate of approximately 71.92%. These are associations and do not establish that support activity causes churn.

Satisfaction was missing for a substantial proportion of records. Observed churn was approximately 66.34% among customers with missing satisfaction and 67.32% among customers with recorded satisfaction, illustrating why missingness should be handled explicitly.

5.3 Visualisation

The project produced seven Matplotlib visualisations and three interactive Plotly HTML visualisations covering overall churn, membership tier, recency, frequency, monetary value, support activity, correlations, channel churn and multivariate relationships.

6. Machine Learning Methodology

6.1 Final feature set

- membership_tier
- preferred_channel
- state
- referral_source
- marketing_opt_in
- Frequency
- Monetary_Total
- Monetary_Average
- Support_Ticket_Count
- Average_Satisfaction
- Average_Resolution_Time

Categorical variables were encoded inside the preprocessing pipeline and numeric variables were passed through the modelling workflow. A fixed random seed was used for reproducibility.

6.2 Models

The project evaluated a majority-class baseline, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Neural Network, Voting Ensemble and tuned tree-based models. Evaluation used Accuracy, Balanced Accuracy, Macro F1, ROC-AUC, precision, recall, confusion matrices and false-positive/false-negative analysis.

6.3 Class imbalance

Because 67.24% of the target population was churned, accuracy alone was not sufficient. Balanced Accuracy, Macro F1, ROC-AUC and class-specific metrics were used to provide a more informative evaluation.

7. Model Results

Model	Accuracy	Churn Precision	Churn Recall	Churn F1	Bal. Acc.	ROC-AUC
Baseline	0.6683	0.6683	1.0000	0.8012	0.5000	0.5000
Neural Network	0.6675	0.6874	0.9214	0.7874	0.5387	0.6469
Gradient Boosting	0.6806	0.7149	0.8686	0.7842	0.5853	0.6588
Random Forest	0.6617	0.6986	0.8686	0.7744	0.5568	0.6393
Voting Ensemble	0.6166	0.7006	0.7445	0.7219	0.5517	0.6260

Model	Accuracy	Churn Precision	Churn Recall	Churn F1	Bal. Acc.	ROC-AUC
Tuned Random Forest	0.6576	0.7575	0.7174	0.7369	0.6273	0.6649
Decision Tree	0.6018	0.6961	0.7174	0.7066	0.5431	0.5430
Logistic Regression	0.6248	0.7629	0.6364	0.6939	0.6189	0.6585
Tuned Decision Tree	0.6207	0.7643	0.6253	0.6878	0.6183	0.6642

7.1 Final model

The tuned Random Forest was selected as the final project model because it provided a stronger balance between churn precision and recall than the untuned Random Forest and materially improved balanced performance. Its ROC-AUC of 0.6649 indicates useful but limited discrimination.

7.2 Hyperparameter tuning

Grid-search cross-validation tested max_depth, min_samples_leaf and n_estimators. The best configuration was max_depth=10, min_samples_leaf=5 and n_estimators=100, with a best cross-validation score of approximately 0.6279.

8. Model Interpretation and Error Analysis

8.1 Important features

The final pipeline's feature importance placed Frequency and Monetary_Total at the top, followed by Monetary_Average and Average_Resolution_Time. This indicates that purchasing behaviour and customer value are important inputs to the model's predictions.

8.2 False positives and false negatives

On the held-out test set, the tuned Random Forest produced 230 false negatives and 187 false positives. A false negative is a churned customer missed by the model; a false positive is an active customer incorrectly flagged as churned.

Gradient Boosting produced fewer false negatives (107) but more false positives (282). Therefore, the appropriate operational model depends on the financial cost ratio. If missing a churner is more expensive, a higher-recall operating strategy may be appropriate; if retention interventions are expensive, a more selective strategy may be appropriate.

8.3 SHAP

SHAP analysis was completed to provide an explainability layer alongside model-level feature importance. Individual and global SHAP outputs were saved in the project's Phase 9 explainability folder.

9. Business Recommendations

- Use the model to prioritise customers for retention outreach rather than to make automatic irreversible decisions.
- Pay particular attention to declining purchasing frequency and customer value when designing retention segments.
- Use support-ticket volume and resolution experience as customer-experience signals for investigation.
- Run controlled retention campaigns to measure whether interventions actually reduce churn.
- Attach explicit financial costs to false positives and false negatives before choosing a production threshold.
- Monitor model performance over time and retrain when customer behaviour or business processes change.

10. API Deployment

The final model was saved as final_churn_model.joblib and exposed through a FastAPI application with a POST /predict endpoint.

A successful API test returned HTTP 200 with prediction=1, churn_label='Churned' and churn_probability=0.6604. Missing required fields and invalid input types returned HTTP 422 validation responses, demonstrating input validation and error handling.

Docker and Streamlit were treated as optional bonus components and were not required for the core examiner deliverables.

11. Limitations

- The 180-day churn definition is a business rule and may not capture every type of disengagement.
- The target is imbalanced, making accuracy potentially misleading.
- ROC-AUC of 0.6649 indicates useful but limited discrimination rather than certainty.
- Support satisfaction has substantial missingness.
- The analysis is observational and does not establish causation.
- A production system should use time-based validation and monitor future-period performance.
- Recency was excluded from modelling because it directly overlaps with the target definition; future work could add independent engagement signals.

12. Conclusion

This project delivers an end-to-end customer-churn investigation for Meridian Retail Group, covering data quality, EDA, RFM analysis, support analysis, predictive modelling, explainability and API deployment.

The results show strong descriptive differences in customer recency and useful associations between support activity and churn. The tuned Random Forest provides a reproducible starting point for retention prioritisation, but deployment should be guided by explicit business costs, campaign testing and ongoing model monitoring.

Appendix A. Core Deliverables

Deliverable	Project location
Business/Technical Report	Final Submission/Meridian_Retail_Group_Business_Technical_Report.pdf
Cleaned datasets	Data/Cleaned Data/
Visualisations	Data/Phase 8 Visualization/
Final ML model	Data/Phase 9 Machine Learning/final_churn_model.joblib
Evaluation results	Data/Phase 9 Machine Learning/06_Machine_Learning/model_evaluation_results.csv
Tuning results	Data/Phase 9 Machine Learning/06_Machine_Learning/hyperparameter_tuning_results.csv
API	Data/Phase 9 Machine Learning/Deployment/
README	README.md
Final presentation	Data/Phase 9 Machine Learning/Meridian_Retail_Group_Customer_Churn_Final_Presentation.pptx